Communication-Efficient Multi-Objective Federated Sequential Recommendation with Adaptive Expert Routing

A Multi-Objective, Efficiency-Aware Transformer Recommender for Federated Deployment

Project Report

Prepared by: **Arvin Samuel A, Aayath Hussain Parvez, Anirudh SK**

VIT Chennai — B.Tech. Computer Science and Engineering

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Contents

1 Problem Statement **2**
- 1.1 Formal Problem Definition & Contributions . . . 2
- 1.2 Research Questions . . . 3

2 Literature Review **3**
- 2.1 Comparative Summary . . . 4
- 2.2 Research Gap . . . 6

3 Proposed Methodology **7**
- 3.1 Base Sequential Recommender . . . 7
- 3.2 Multi-Objective Training Signal . . . 7
- 3.3 Resolving Multi-Objective Gradient Interference . . . 8
  - 3.3.1 Adaptive Routing Vector Mixture-of-Experts (ARV-MMoE) . . . 8
  - 3.3.2 Inference-Time Objective Embedding . . . 9
  - 3.3.3 Dynamic Loss Balancing (GradNorm) . . . 9
- 3.4 Systems-Level Design for Deployment . . . 10
  - 3.4.1 Efficient Candidate Retrieval . . . 10
  - 3.4.2 Federated Personalization with LoRA . . . 10
- 3.5 Evaluation Protocol . . . 10
- 3.6 Ablation Studies . . . 11
- 3.7 Diagnostics . . . 12
  - 3.7.1 Routing Diagnostics . . . 12
  - 3.7.2 Gradient-Interference Diagnostics . . . 13

4 System Architecture **14**
- 4.1 Data Pipeline . . . 14
- 4.2 Core Model Architecture . . . 14
- 4.3 Serving / Inference Path . . . 14
- 4.4 Federated Learning Architecture . . . 14
- 4.5 Evaluation and Monitoring Subsystem . . . 15
- 4.6 End-to-End System Diagram (Textual) . . . 16

1 Problem Statement

Modern sequential recommendation systems are expected to satisfy three objectives that are frequently in tension with one another: (i) high ranking quality, which typically demands large-capacity models trained on rich, centrally pooled interaction histories; (ii) data locality, driven by regulatory pressure and user expectations that raw interaction data should not leave the device; and (iii) practical deployability, which requires low inference latency, a small memory footprint, and bounded communication cost when models are trained or personalized across many clients. Centralized transformer-based recommenders (e.g., SASRec, BERT4Rec) achieve strong ranking quality but assume unrestricted access to a global interaction log, an assumption that is increasingly untenable in consumer applications.

As surveyed in Section 2, the federated recommendation literature has largely optimized these objectives in isolation. Works such as Reddy et al. and Lee and Cho demonstrate that Transformer-based sequential models can be federated with reasonable accuracy, but they do not jointly reason about deployment cost. A large body of privacy-preserving work shows that formal guarantees (e.g., differential privacy) can be layered onto federated recommenders, but consistently reports that the added protection mechanisms increase computational or communication overhead, and in the DP case, directly trade off against accuracy. None of the surveyed works simultaneously (a) train on multiple complementary sequential objectives without gradient interference, (b) explicitly measure and report the systems-level cost of deployment (latency, model size, communication bytes/round), and (c) validate, through controlled ablations, that observed gains are attributable to specific architectural choices.

A second, largely unaddressed problem in the surveyed literature is multi-objective interference. Combining next-item prediction with auxiliary objectives such as masked item modeling, multi-step forecasting, and contrastive learning is known to improve representation quality, but naively summing multiple objectives can produce conflicting gradients and imbalanced optimization dynamics: objectives with larger gradient magnitude may dominate updates, objectives evolve at different rates, and joint training can degrade below the performance of any single objective trained alone. This failure mode is not addressed by any of the twenty reviewed papers.

1.1 Formal Problem Definition & Contributions

Given a set of clients $\mathbb{N} = \{1, \ldots, N\}$, each holding a private, non-IID sequence of item interactions S_i for its users, the objective is to learn a sequential recommendation model f_θ that maximizes ranking quality while minimizing deployment cost. To achieve this, three core scientific contributions are explicitly outlined:

Contribution 1: **Objective-conditioned ARV-MMoE:** Dynamically allocate shared expert capacity according to the user's sequential representation and learning objective. The routing mechanism is parameterized through a shared router conditioned on objective embeddings, replacing the conventional approach of independent, task-specific gating functions.

Contribution 2: **ARV + GradNorm for multi-objective learning:** Combine adaptive routing with differentiated expert pathways (ARV-MMoE) and dynamic objective balancing (GradNorm) to mitigate or redistribute multi-objective interference at both the representation and optimization levels.

Contribution 3: **Federated LoRA adaptation:** Enable parameter-efficient local adaptation on each client and lightweight federated updates to make the proposed multi-objective recommender practical under communication constraints.

1.2 Research Questions

This work explicitly answers the following research questions through controlled systems evaluation:

RQ1: Does multi-objective training improve sequential recommendation compared with a single-objective Transformer?

RQ2: Does ARV-MMoE outperform conventional task-specific MMoE by conditioning expert routing jointly on user sequence context and learning objective?

RQ3: Does combining ARV-MMoE with GradNorm reduce objective interference and improve multi-objective optimization?

RQ4: Can federated LoRA preserve the recommendation gains of the proposed architecture while substantially reducing communication cost?

RQ5: What are the trade-offs between ranking quality, inference latency, model size, memory usage, and federated communication cost?

Each research question maps to specific ablation experiments:

Table 1: Research question to experiment mapping.

RQ	Experiment	Ablation
RQ1	Single-objective vs. multi-objective Transformer	Architecture progression (1, 2)
RQ2	Conventional MMoE vs. ARV-MMoE	MMoE vs. ARV-MMoE; Routing formulation
RQ3	ARV vs. GradNorm vs. combined	ARV vs. GradNorm; Gradient-interference diagnostics
RQ4	Global vs. FedLoRA vs. FedLoRA + ARV	Personalization; Federated efficiency
RQ5	Full system evaluation across all dimensions	All ablations; Diagnostics; System ablations

2 Literature Review

This review covers twenty recent works (2024–2026) spanning federated sequential recommendation, privacy-preserving mechanisms, graph-based federated architectures, personalization strategies for non-IID clients, and communication-efficient federated protocols. The five studies with the most direct bearing on the proposed methodology are discussed individually below; the remaining fifteen are then grouped thematically into three clusters: (i) differential privacy, (ii) graph-based federated architectures, and (iii) personalization and client contribution. Foundational references for the multi-objective and gradient-balancing methods central to this proposal are cited separately: MMoE [22] and GradNorm [23].

Reddy et al. [1] provide the most direct architectural precedent for this proposal, combining federated learning with Transformer-based sequential recommendation. They implement federated BERT4Rec and a modified Transformer-XL variant (BST) and evaluate on the Amazon and MovieLens-1M datasets, reporting approximately 87.4% accuracy for federated BERT4Rec on MovieLens-1M and 0.80 MAE for federated BST on Amazon. Their results confirm that causal/masked Transformer encoders retain most of their centralized-training accuracy once

federated, motivating the causal Transformer backbone adopted in Section 3.1. However, the authors explicitly flag communication efficiency and heterogeneous client data as open problems — a gap this proposal targets directly through parameter-efficient communication and weighted aggregation.

Lee and Cho [5] address a closely related but distinct problem: slow convergence in federated sequential recommenders caused by short, heterogeneous local interaction histories. They propose six local sequence-augmentation strategies — Random Push, Self Push, Cut, Delete, Replace, and Shuffle — and show that techniques producing greater sequence-length variation most effectively reduce the number of federated rounds required. This proposal adopts the same six strategies directly, using them to construct the augmented views consumed by the sequence-level contrastive objective in Section 3.2.

Liu et al. [16] propose FedBPG, which separates a shared global backbone from a lightweight, locally trained personalized component, using knowledge distillation rather than direct parameter averaging to transfer common knowledge from participating clients. The architecture is the closest existing analogue to the global-backbone-plus-client-adapter design adopted in Section 3.4.2, differing mainly in mechanism: this proposal personalizes through low-rank LoRA adapters and adaptive routing rather than a distilled personalized sub-network.

Kong et al. [13] target the communication overhead of repeated federated rounds directly, proposing Oh-FedRec, a one-shot vertical federated recommender that trains in a single communication round by removing the requirement that all parties share an identical model structure. This motivates, by contrast, the iterative-but-lightweight protocol adopted in Section 3.4.2: rather than eliminating rounds entirely, this proposal keeps them iterative but keeps each round's payload small.

Yang et al. [20] propose a hierarchical federated recommendation framework that models user preferences at the individual, cluster, and global levels simultaneously, combined with a dynamic, quality-aware aggregation mechanism. This directly informs the cold-start evaluation protocol adopted in Section 3.5, and motivates reporting personalization gains broken out by cold-start status rather than only in aggregate.

A second cluster of work applies formal differential privacy to the recommendation pipeline. Lin et al. [2] propose PriCDSR, adding calibrated noise to masked items in a cross-domain sequential recommender under (ϵ, δ)-differential privacy. Chen et al. [17] apply a quantized-projection local differential privacy mechanism to graph representations in FedHGPP. Batmaz et al. [18] extend local DP to multi-criteria collaborative filtering. All three consistently report the same privacy–utility trade-off, reinforcing the decision in this proposal to rely on strict data locality and parameter-efficient updates rather than injecting DP noise.

A larger cluster augments federated recommendation with graph structure ([3], [6], [7], [9], [10], [11], [12], [15]), relying on decentralized client-graphs, homomorphic encryption, or functional encryption. Across this cluster, graph-based federated recommenders consistently trade accuracy gains for substantial systems complexity, a trade-off the proposed Transformer-based design avoids by modeling sequences through attention rather than explicit graph construction.

The remaining studies address personalization, client contribution, and efficiency from other angles, including clustering [4], Shapley-inspired client contribution scoring [8], trusted-node architectures [19], and entropy-guided trees [14].

2.1 Comparative Summary

Table 2: Comparative summary of the twenty reviewed studies (2024–2026).

#	**Study**	**Core Technique**	**Reported Result**	**Key Limitation**
1	Reddy et al. (2024, IEEE Access)	Federated BERT4Rec / BST (Transformer-XL variant)	∼87.4% acc. (ML-1M, BERT4Rec); 0.80 MAE (Amazon, BST)	Communication efficiency and client heterogeneity unresolved
2	Lin et al. (2024, TKDE)	PriCDSR — differential privacy, cross-domain sequential rec.	Outperforms single-domain BERT4Rec across $\epsilon = 10/5/2$	Accuracy degrades as privacy budget tightens
3	Li et al. (2024, ACM TIST)	DFedRec — decentralized client-graph, matrix factorization	Comparable to centralized baselines	Graph construction adds coordination/scalability overhead
4	Xu et al. (2024, Applied Soft Computing)	User clustering + meta-learning	Improved effectiveness and communication efficiency	Added computational complexity from clustering/adaptation
5	Lee & Cho (2025, IEEE Access)	Sequence augmentation (6 strategies) for federated seq. rec.	Faster convergence, better cold-user performance	Effectiveness strategy-dependent
6	Wang et al. (2025, IEEE TCSS)	FedPCL — GNN + contrastive learning, cluster/global models	Outperforms baselines on Recall/NDCG (5 datasets)	GNN + contrastive + clustering increases system complexity
7	Chen et al. (2025, IEEE TSC)	DeFedGCN — decentralized GCN, local DP	Up to 7.4% accuracy gain over non-DoS-protected baselines	P2P + graph expansion overhead
8	Yin et al. (2025, IEEE TCE)	CA-PFL — Shapley-inspired contribution scoring	Improved accuracy and training efficiency vs. FedAvg	Contribution scoring adds computation
9	Gu et al. (2025, IEEE TCSS)	Social + knowledge graph, GAT, homomorphic encryption	+2.1% AUC, +3.3% F1 vs. single-side-info baselines	HE + graph processing raises complexity
10	Pu et al. (2025, IoT-J)	Federated GCN + DistilBERT for course recommendation	Substantially outperforms conventional baselines	GCN + HE implementation complexity

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#	Study	Core Technique	Reported Result	Key Limitation
11	Zhang et al. (2025, TPAMI)	PRGNN — subgraph split, functional encryption, HAM/CSM	+22.1% training efficiency; −22% communication	Architectural/computational overhead from encryption
12	Hu et al. (2025, TNSE)	P^2FedRec — user–user & user–item relationship graphs	More personalized than conventional FedRec	Encrypted graph exchange raises comm./compute cost
13	Kong et al. (2025, IEEE TCE)	Oh-FedRec — one-shot vertical federated learning	Matches/exceeds iterative baselines, far lower comm. cost	Sensitive to initial model quality
14	Jha & Tripathi (2025, IEEE Wireless Networks)	Entropy-guided π-tree for IoT sensor recommendation	Outperforms baselines on sensor data	Not federated / not privacy-focused
15	Lou et al. (2025, Scientific Reports)	FedGA — divergence-resistant aggregation for GNN session rec.	Competitive with centralized models under non-IID data	Added optimization complexity; session-specific
16	Liu et al. (2026, IEEE TCE)	FedBPG — global backbone + personalized component, distillation	Better generalization/personalization under non-IID data	Distillation increases compute cost
17	Chen et al. (2026, IEEE TIFS)	FedHGPP — quantized-projection local DP for heterogeneous graphs	HR@10 $\approx$ 0.476, NDCG@10 $\approx$ 0.254 at $\epsilon = 2$	Privacy noise still reduces accuracy
18	Batmaz et al. (2026, Expert Systems)	LocalDP-MCCF — LDP for multi-criteria ratings	Reasonable accuracy retained under LDP	Inherent privacy–utility trade-off
19	Yan et al. (2026, Expert Systems)	FastPFRec — trusted-node GNN + secure sharing	+8.1% accuracy, −34.1% convergence time	Trusted-node architecture adds system complexity
20	Yang et al. (2026, Physica A)	Hierarchical multi-level personalization + dynamic aggregation	Gains in HR@10/NDCG@10, esp. cold-start	Hierarchical DP mechanisms need extra tuning

2.2 Research Gap

The reviewed literature separately explores federated sequential recommendation, multi-objective learning, expert-based architectures, personalization, and communication-efficient adaptation. Foundational work on MMoE [22] establishes the shared-expert multi-objective paradigm with task-specific gating, while GradNorm [23] provides gradient-balanced normalization for multi-objective optimization. The reviewed literature does not explicitly evaluate the proposed com-

bination of objective-conditioned adaptive expert routing and gradient-aware multi-objective optimization within a communication-constrained federated sequential recommendation setting. The proposed work investigates this specific combination: whether shared-task-conditioned parameterization of MMoE-style routing, combined with gradient-aware balancing, yields measurable improvements over independently studied components within a federated deployment context.

3 Proposed Methodology

The central contribution of this system is an objective-conditioned adaptive routing mechanism that dynamically allocates shared expert capacity across multiple sequential recommendation objectives. GradNorm complements this mechanism by balancing their optimization contributions, while LoRA-based federated personalization enables the resulting model to adapt to heterogeneous clients with strictly limited communication.

3.1 Base Sequential Recommender

The backbone is a causal Transformer encoder over each user's chronological interaction sequence (max length 50). Item and position embeddings are learnable; the output projection layer is tied to the item embedding matrix, and training uses sampled softmax with post-hoc temperature-scaled logit calibration at evaluation time. Every attention call applies a causal mask and a key-padding mask simultaneously. Item metadata (genre, and top-k genome tag relevance scores) is fused into the input embedding to support cold-item generalization.

3.2 Multi-Objective Training Signal

Four complementary objectives are combined during pretraining and personalization. Each addresses a distinct operational problem in sequential recommendation:

- **Next-item prediction (L_1, $t = 1$) — Immediate next-item relevance.** The primary causal-language-modeling-style objective: predict the next item given the full interaction history up to time t. Formally, $L_{\text{next}} = \mathcal{L}(y_{t+1} \mid H_{\leq t})$. This objective directly corresponds to the deployed recommendation task and is especially important for immediate recommendation quality.

- **Masked item modeling (L_2, $t = 2$) — Robust contextual reconstruction.** A BERT-style denoising objective that regularizes the encoder by reconstructing masked items from surrounding context; the mask-token index is allocated outside the real item ID range. This addresses incomplete or noisy interaction histories by forcing the encoder to learn robust contextual representations.

- **Multi-step forecasting (L_3, $t = 3$) — Longer-horizon intent forecasting.** Predict the next k items at horizons $j = 2, \ldots, k+1$, enforcing causality so future context does not leak. Formally, $L_{\text{forecast}} = \frac{1}{k} \sum_{j=2}^{k+1} \mathcal{L}(y_{t+j} \mid H_{\leq t})$, with $k = 3$ as the default. This models evolving, longer-horizon user intent rather than only the immediate next interaction.

- **Contrastive learning (L_4, $t = 4$) — Sequence representation invariance.** A sequence-level contrastive loss over augmented views of the same user sequence, computed on a subsampled interval of batches. This improves robustness to behavioral perturbation and sparse, non-IID sequences by encouraging the encoder to produce consistent representations under augmentation.

Next-item prediction is retained as a dedicated objective because it directly corresponds to the deployed recommendation task, whereas multi-step forecasting models longer-horizon user intent. The serving path uses $e_{\text{rec}} \equiv e_{\text{next}}$, reinforcing that next-item relevance is the primary deployment goal.

3.3 Resolving Multi-Objective Gradient Interference

Naively summing these four losses causes gradient interference. This proposal investigates interference at two complementary levels: representational routing and optimization balancing.

3.3.1 Adaptive Routing Vector Mixture-of-Experts (ARV-MMoE)

It is critical to distinguish between the structural architecture (MMoE) and the routing formulation (ARV). MMoE itself is not claimed as novel; it is a well-established shared-expert architecture for multi-objective learning [22]. The central contribution of this work is the *adaptive routing vector* (ARV) formulation, which replaces the conventional gating mechanism of MMoE with a shared, objective-conditioned router.

Conventional MMoE. In the conventional MMoE baseline, each objective t is served by an *independently parameterized* gating function:

$$Z_t^{MMoE} = \sum_k g_{t,k}(H_u) E_k(H_u), \tag{1}$$

where the gating function for objective t is:

$$g_t(H_u) = \text{softmax}(W_t H_u + b_t). \tag{2}$$

Each objective t has its own parameters (W_t, b_t), meaning that no information is shared *across routing policies*—each gate is learned in isolation from the others. The gate for objective t maps a hidden representation of dimension d to K expert weights, so $W_t \in \mathbb{R}^{K \times d}$ and $b_t \in \mathbb{R}^K$. The total number of trainable gate parameters is $T(Kd + K)$, where T is the number of objectives, d is the hidden dimension, and K is the number of experts.

ARV-MMoE: shared objective-conditioned routing. ARV-MMoE replaces the independent gates with a *single shared router* parameterized by (W_r, b_r) and conditioned on an objective embedding e_t:

$$r_{u,t} = \text{softmax}\left(W_r[H_u; e_t] + b_r\right), \tag{3}$$

where $r_{u,t} \in \mathbb{R}^K$ is the routing distribution over K shared experts for user u and objective t. The objective-specific representation is then:

$$Z_{u,t} = \sum_{k=1}^{K} r_{u,t,k} E_k(H_u). \tag{4}$$

The critical design choice is that (W_r, b_r) are *shared* across all objectives; task-specificity is introduced solely through the embedding e_t. If $H_u \in \mathbb{R}^d$ and $e_t \in \mathbb{R}^{d_{\text{emb}}}$, the shared router has $K(d + d_{\text{emb}}) + K$ parameters, and the T objective embeddings add $T \cdot d_{\text{emb}}$ parameters. The ARV-MMoE total is therefore $K(d + d_{\text{emb}}) + K + T \cdot d_{\text{emb}}$, compared to the MMoE baseline's $T(Kd+K)$. The hypothesis is that this parameter-sharing/factorization structure enables information sharing across routing policies, yielding more sample-efficient routing than independent task-specific gates. Total trainable parameter count is matched as closely as practical, while router, expert, and total parameter counts are reported. Where exact matching is not possible, comparisons are conducted across capacity-matched configurations.

User-adaptive routing without persistent personalization vectors. Because H_u is the Transformer representation of user u's interaction sequence, H_u is inherently user-specific. Consequently, $r_{u,t} = f(H_u, e_t)$ naturally produces different routing distributions for different users: for users u and v, $H_u \neq H_v \implies r_{u,t} \neq r_{v,t}$ in general. The routing is *context-adaptive* by construction—it is a function of the user's sequence representation and the objective embedding—and no separate persistent client-specific routing vector or personalization state is required. This keeps the architecture focused on the primary research hypothesis.

Research question. The central research question addressed by this design is whether a *shared* objective-conditioned routing formulation—in which routing policies share the same router parameters (W_r, b_r) and differ only through their objective embeddings e_t—provides a measurable advantage over the conventional approach of independently parameterized, task-specific gating functions $g_t(H_u)$.

3.3.2 Inference-Time Objective Embedding

The routing formulation above is defined over training objectives $t \in \{1, 2, 3, 4\}$, each with its own learnable embedding e_t. At serving time, no training objective is simultaneously active; the system performs a single recommendation pass conditioned on next-item ranking. The inference objective embedding is therefore defined as:

$$e_{\text{rec}} \equiv e_{\text{next}}. \tag{5}$$

The routing vector used at inference is:

$$\boxed{r_u^{\text{rec}} = \text{softmax}\left(W_r[H_u; e_{\text{rec}}] + b_r\right)} \tag{6}$$

and the final representation fed to candidate scoring is:

$$Z_u^{\text{rec}} = \sum_{k=1}^{K} r_{u,k}^{\text{rec}} E_k(H_u). \tag{7}$$

This choice is both simpler and more defensible: the deployed recommendation task is next-item ranking, so the inference routing embedding is identified with the next-item training embedding e_{next}. Because e_{next} already receives gradients through the next-item loss during training, e_{rec} is trained by construction—there is no untrained, inference-only embedding. This eliminates any ambiguity about how the inference routing embedding is learned and ensures that the routing policy at serving time is exactly the routing policy learned for the next-item objective.

3.3.3 Dynamic Loss Balancing (GradNorm)

Define the total multi-objective loss as:

$$L = \sum_t \lambda_t L_t \tag{8}$$

GradNorm dynamically adjusts λ_t according to objective training dynamics and gradient magnitudes. The core hypothesis relies on a central explanatory framework:

- **ARV-MMoE** → Which experts should process this objective?
- **GradNorm** → How strongly should each objective train the model?
- **LoRA** → How can clients adapt the model with low communication?

These mechanisms address complementary issues: ARV-MMoE dynamically allocates expert capacity among objectives, while GradNorm dynamically balances the optimization contribution of those objectives.

3.4 Systems-Level Design for Deployment

3.4.1 Efficient Candidate Retrieval

Full-catalogue scoring at inference time is replaced with an ANN-based candidate generation stage, using FAISS (CUDA) or a brute-force matrix-multiply fallback (MPS). Ground-truth items are force-included in the candidate set during evaluation to prevent tail items from being structurally excluded.

3.4.2 Federated Personalization with LoRA

The system strictly separates global and local parameters to enforce communication efficiency:

Parameter Category	Components
Global / Shared	Transformer backbone, shared experts, ARV router parameters (W_r, b_r), objective embeddings (e_t), shared recommendation heads
Local / Client-Specific	LoRA adapters

During federated personalization, the shared backbone remains frozen. The raw interaction history remains entirely local, reducing direct exposure of user histories. The federated communication payload consists strictly of the LoRA parameter updates ΔW_u^{LoRA} from each client; no raw routing vectors, no interaction data, and no backbone parameters are transmitted.

The frozen global backbone parameters are denoted as W_{base}. For client u, the locally adapted model is:

$$W_u = W_{\text{base}} + \Delta W_u^{\text{LoRA}}, \tag{9}$$

where the LoRA update is parameterized as a low-rank decomposition:

$$\Delta W_u^{\text{LoRA}} = B_u A_u. \tag{10}$$

The backbone W_{base} remains fixed across all federated rounds.

The server aggregates LoRA parameter updates from participating clients using weighted federated averaging:

$$\boxed{\Delta W_{\text{global}}^{\text{LoRA}} = \sum_{u \in \mathcal{S}_r} w_u \, \Delta W_u^{\text{LoRA}}} \tag{11}$$

where $\mathcal{S}_r$ is the set of clients sampled in round r, and $w_u = \frac{n_u}{\sum_{v \in \mathcal{S}_r} n_v}$ is the weight for client u based on its local sample count n_u. The server aggregates *LoRA parameter updates*—not raw routing vectors, not interaction data, and not backbone parameters. The deployed global adapted model is:

$$\boxed{W_{\text{deploy}} = W_{\text{base}} + \Delta W_{\text{global}}^{\text{LoRA}}} \tag{12}$$

This design keeps W_{base} fixed across federated rounds. The shared ARV router and objective embeddings remain global parameters, while client-specific adaptation is represented exclusively through LoRA updates. The final deployed model is the sum of the frozen backbone and the globally aggregated LoRA adapter.

Federated training keeps raw interaction data on the client while transmitting only lightweight LoRA parameter updates. The model therefore operates with a global shared representation, adaptive per-sequence routing (ARV-MMoE), and local LoRA-based federated client adaptation.

3.5 Evaluation Protocol

The system is evaluated across the following critical dimensions:

Table 3: Multi-dimensional evaluation protocol.

Dimension	Metric(s)
Recommendation quality	NDCG@10, HR@10, cold-start performance, long-tail performance
Optimization	Per-objective losses, gradient norms, gradient cosine similarities, convergence rounds
Inference	Latency (ms/query), p95 latency, model size (MB), memory
Federated efficiency	Upload bytes/round, download bytes/round, total communication until convergence, number of rounds
Federated adaptation	Global model vs FedLoRA performance, communication cost, convergence rounds
Adaptive routing diagnostics	Routing distributions, routing entropy, expert utilization, objective-specific routing differences

3.6 Ablation Studies

The ablations directly map to the research questions defined in Section 1.2 and are organized into seven categories.

A. **Architecture progression:** Evaluates the full model-building stack in order:

   1. Transformer + next-item prediction (single-objective baseline)
   2. Transformer + all four objectives (multi-objective without routing)
   3. Transformer + conventional task-specific MMoE
   4. Transformer + ARV-MMoE
   5. Transformer + ARV-MMoE + GradNorm
   6. Federated ARV-MMoE + GradNorm + LoRA

B. **Routing formulation:** Compares the input-only adaptive router $r = f(H_u)$ against the proposed objective-conditioned router $r = f(H_u, e_t)$. This tests whether conditioning routing on the objective embedding provides a measurable advantage over routing conditioned solely on the user representation.

C. **MMoE vs ARV-MMoE:** Directly compares conventional task-specific MMoE $g_t(H_u)$ against the proposed ARV-MMoE $r_{u,t} = f(H_u, e_t)$. The conventional MMoE baseline is parameter-matched as closely as practical to ARV-MMoE. The purpose of parameter-matching is to ensure that any observed performance differences are attributable to the routing parameterization and objective conditioning rather than to differences in parameter count or model capacity. If exact parameter matching is not achievable for a given implementation, the comparison must report the following quantities to allow a fair assessment: (i) total trainable parameters, (ii) expert parameters, (iii) router/gating parameters, and (iv) inference latency.

D. **ARV vs GradNorm:** Evaluates four configurations: (i) neither ARV-MMoE nor GradNorm (multi-objective Transformer); (ii) ARV-MMoE only; (iii) GradNorm only; (iv) ARV-MMoE + GradNorm. This tests whether the two mechanisms mitigate or redistribute complementary sources of interference.

E. **Personalization:** Compares the global model, global + LoRA, and global + LoRA + ARV (context-adaptive routing with LoRA adaptation). The ARV mechanism is context-adaptive

expert routing (not a persistent client-specific personalization state); LoRA is the explicit federated client-adaptation mechanism.

F. **Objective ablation:** Removes each objective individually (next-item, MIM, multi-step forecasting, contrastive learning) to assess each objective's contribution to overall recommendation quality.

G. **Expert-count ablation:** Evaluates $K \in \{2, 4, 6, 8\}$ to determine the effect of the number of shared experts on recommendation quality and inference efficiency.

H. **Existing system ablations:** Preserves the following system-level experiments: metadata integration, sequence length, cold-start performance, long-tail performance, latency analysis, and communication cost analysis.

3.7 Diagnostics

3.7.1 Routing Diagnostics

To measure whether ARV actually behaves adaptively and to test for meaningful expert specialization, the following diagnostics are reported. Routing is not claimed to be specialized unless the experiments demonstrate it.

Routing entropy.

$$\mathcal{H}(r) = -\sum_{k} r_k \log r_k \tag{13}$$

Low entropy indicates concentrated routing (one or few experts dominate); high entropy indicates diffuse routing. Mean entropy is reported across all user sequences and per-objective, and compared across model variants.

Expert utilization.

$$U_k = \frac{1}{N} \sum_{i=1}^{N} r_{i,k} \tag{14}$$

This measures the fraction of total routing mass assigned to expert k. Uniform utilization (all $U_k \approx 1/K$) is a warning sign of routing collapse; a single expert with $U_k \gg 1/K$ indicates concentration.

Objective-specific routing differences. For each objective $t \in \{\text{next-item, MIM, forecast, contrast}\}$, the mean routing distribution $\bar{r}_t$ across all user sequences is computed. Pairwise cosine similarity and KL divergence between objective routing distributions are then reported:

$$\text{sim}_{t,t'} = \cos(\bar{r}_t, \bar{r}_{t'}), \quad \text{KL}_{t,t'} = \sum_{k} \bar{r}_{t,k} \log \frac{\bar{r}_{t,k}}{\bar{r}_{t',k}}. \tag{15}$$

Low cosine similarity and high KL divergence between objectives indicate that the routing mechanism differentiates expert allocation across objectives. High similarity and low divergence suggest the routing is not truly objective-adaptive.

Objective $\times$ expert heatmap. A heatmap of mean routing weights $\bar{r}_{t,k}$ is constructed with rows indexed by objective t and columns by expert k. This visual diagnostic reveals whether specific experts are preferentially used by specific objectives and whether the routing distribution is meaningfully non-uniform.

Collapse tests. Two failure modes are explicitly tested for using the following diagnostic thresholds, which are selected for analysis and should be interpreted alongside the continuous distributions:

- **Uniform routing collapse:** All $U_k \approx 1/K$ within a tolerance (e.g., coefficient of variation < 0.1). This indicates the router is not differentiating experts.
- **Single-expert collapse:** One expert captures $> 80\%$ of total routing mass ($U_k > 0.8$ for some k). This indicates the router is collapsing to a single expert.

These diagnostics are used to determine whether routing is meaningfully adaptive and whether pathological collapse occurs. Low-entropy routing is not claimed to be inherently undesirable: a router that concentrates on one or few experts may be legitimate if certain objectives naturally specialize heavily. The continuous distributions of routing entropy and expert utilization provide the primary evidence; the thresholds serve as secondary indicators.

3.7.2 Gradient-Interference Diagnostics

Gradient-interference diagnostics test whether the proposed mechanisms meaningfully affect the distribution of conflicting gradients and whether ARV and GradNorm mitigate or redistribute complementary sources of multi-objective interference.

Gradient cosine similarity. For each objective pair (i, j), the following is computed:

$$C_{ij} = \cos(\nabla L_i, \nabla L_j) \tag{16}$$

Negative cosine similarity is direct evidence of conflicting gradients: the two objectives push the parameters in opposite directions. Non-negative cosine similarity does not prove the complete absence of interference — it merely indicates that the gradients are not locally opposed in the direction being measured. The full pairwise matrix and the mean off-diagonal cosine similarity are reported for each model variant.

Per-objective gradient norms. For each objective t, the mean gradient norm $\|\nabla L_t\|$ across training steps is reported. Large disparities in gradient norms across objectives indicate that some objectives dominate the update direction, which is a form of interference even when cosine similarities are non-negative. The coefficient of variation across objectives is reported as a scalar summary of norm imbalance.

Convergence behavior. Per-objective loss curves and overall recommendation metrics (NDCG@10, HR@10) are tracked over training steps. Whether objective losses converge at comparable rates and whether the ranking metrics improve monotonically or exhibit instability are assessed. Diverging or oscillating loss curves are evidence of unresolved interference.

Comparison across model variants. The multi-objective Transformer, conventional MMoE, ARV-MMoE, and ARV-MMoE + GradNorm are compared. ARV-MMoE is not claimed to eliminate gradient interference; rather, the hypothesis is that ARV-MMoE reduces or redistributes conflicting gradient interactions by providing differentiated expert pathways for each objective. GradNorm, independently, is hypothesized to balance optimization contributions by dynamically adjusting per-objective loss weights, thereby mitigating magnitude-based interference. The combination is hypothesized to mitigate or redistribute both representation-level and magnitude-level sources of interference simultaneously.

4 System Architecture

The system is organized into five subsystems: a data pipeline, the core model architecture, a candidate-retrieval / serving layer, a federated learning layer, and an evaluation / monitoring layer.

4.1 Data Pipeline

Raw MovieLens-20M tables are validated at startup and processed into per-user chronological interaction sequences. Sequence augmentation (Random Push / Self Push / Cut / Delete / Replace / Shuffle) and metadata fusion (top-k genome tags) are pre-computed and pre-tensorized.

Dataflow: *Raw CSVs → preprocessing.py → sequence_ generator.py → augmentation.py → pre-tensorized dataset objects.*

4.2 Core Model Architecture

Table 4: Core model components and their implementation modules.

Layer / Module	Function	Module
Embedding layer	Tied item embedding + learnable positional embedding + metadata fusion	`transformer.py`
Transformer encoder blocks	Causal + key-padding masked multi-head self-attention	`transformer.py`
ARV-MMoE Router	Objective-conditioned adaptive expert routing ($r_{u,t} = \text{softmax}(W_r[H_u; e_t] + b_r)$); inference uses $e_{\text{rec}} \equiv e_{\text{next}}$	`transformer.py`
Task heads	Next-item (tied output), MIM, multi-step (k offset), contrastive projection	`transformer.py`, `losses.py`
GradNorm controller	Dynamic per-objective loss weighting for multi-objective optimization	`losses.py`
LoRA adapters	Communication-efficient client personalization	`lora.py`
ANN / retrieval stage	FAISS (CUDA) or brute-force matmul (MPS) candidate generation	`ann_retrieval.py`

4.3 Serving / Inference Path

At serving time, the system performs a single recommendation pass using the inference routing embedding $e_{\text{rec}} \equiv e_{\text{next}}$. The dataflow is: **Dataflow:** *User sequence → embedding + Transformer → shared representation H_u → ARV routing $r_u^{\text{rec}} = \text{softmax}(W_r[H_u; e_{\text{rec}}] + b_r)$ → expert mixture $Z_u^{\text{rec}} = \sum_k r_{u,k}^{\text{rec}} E_k(H_u)$ → ANN candidate generation → calibrated scoring using Z_u^{rec} → ranked top-N.*

4.4 Federated Learning Architecture

Dataflow (per round): *Server broadcasts W_{base} to clients → client performs local training on private sequence data → client uploads LoRA update ΔW_u^{LoRA} → server aggregates $\Delta W_{\text{global}}^{\text{LoRA}} = \sum_u w_u \Delta W_u^{\text{LoRA}}$ → upload/download bytes recorded separately.*

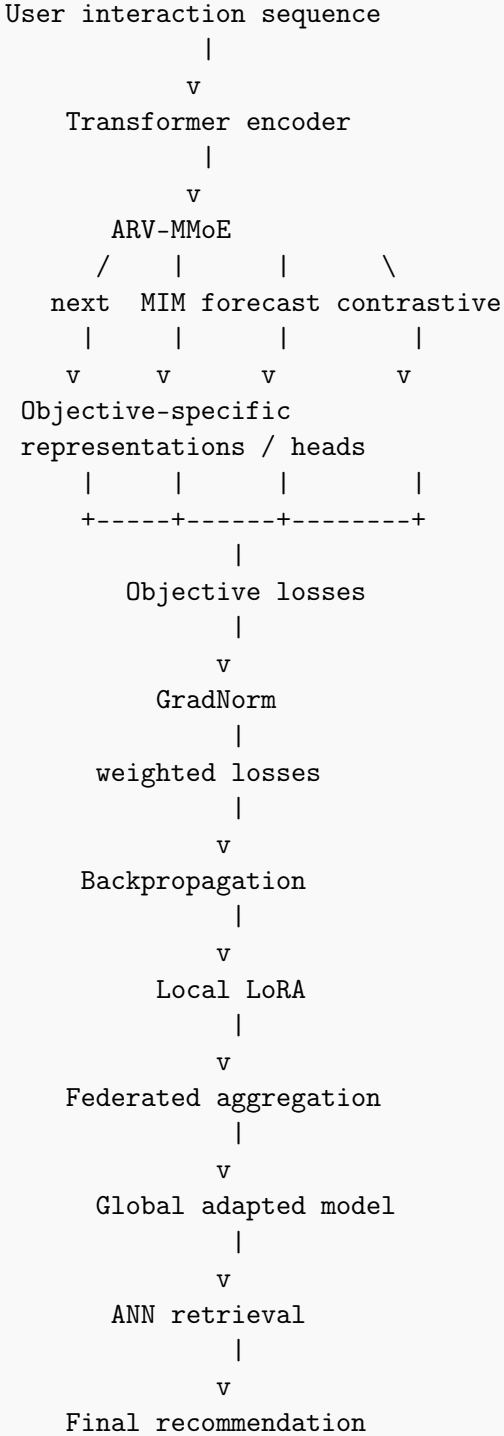
This architecture strictly isolates local representations from the aggregation payload, addressing the inconsistency between local computation and global exchange.

4.5 Evaluation and Monitoring Subsystem

A dedicated `evaluation/` package computes ranking metrics, efficiency metrics, cold-start metrics, and long-tail metrics. Plots (accuracy vs. model size / communication / FLOPs, federated convergence, global-vs-personalized scatter, etc.) are generated.

4.6 End-to-End System Diagram (Textual)

Figure 1. End-to-end system architecture



Note: During training, all four routing paths (r_next, r_MIM, r_forecast, r_contrast) are active. During inference/serving, a single routing pass uses e_rec e_next.
GradNorm operates on objective losses/gradients during training; it is not a forward-pass layer.

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